

CS3 Rubric – Case Study Create

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DS 4002 – Spring 2026 - Instructors: Javier Rasero; TA: Ethan Meidinger

Submission format: Upload link to github repo to canvas

Individual Assignment

General Description: Submit your reproduced results from this project.

Preparatory Assignments: Everything in the course.

Why am I doing this? This is your opportunity to reproduce the project of one of your peers and experiment with timeseries comparison. This specific project runs an STL decomposition model on time series data and evaluates significant differences with a Mann Whitney U test. It is also your chance to learn new skills. This report has been made to be reproduced.

What am I going to do? You will reproduce the result from a student's DS4002 project using time series data concerning PM2.5 (fine particle air quality) patterns across community districts in NY with different populations – one group considered “most” urban and the other “least”. This project employs STL decomposition models as a time series component comparison run multiple times and a Mann Whitney U Test to determine if the results were statistically significant. By reproducing this, you will practice cleaning the data, creating exploratory plots, running evaluation models, and comparatively analyzing the results. Deliverables and resources include:

- variousMaterials - this folder contains everything required for reproducing the results.
 - Data - the .csv file containing the data the project explores pre and post processing
 - Scripts - this folder contains the scripts that must be run in order and require the data found in the Data folder. They can be run in whatever workspace is best suited for you.
 - Sources - this is the background information supporting the motivation and importance of PM2.5 patterns across NYC population density and how they relate to health. There is one academic article and one blog post.
 - References - the list of IEEE references used in this project
- Hook – the one page write up call-to-action for the student that explains the motivation and its importance.
- Rubric – the guidelines necessary to complete this project and reproduce results
- LICENSE and README – guiding documents including this repository's contents, instructions, and rights.

Results produced in whatever environment you choose will satisfy as the final form of output.

Tips for success:

- Stay organized. Follow the reproduction instructions and get to know the GitHub

- Talk to the instructors. If you need help, you may reach out to your classmates.

How will I know I have Succeeded? You will reproduce similar results to the original tests as documented by outputs in the OUTPUT folder. Listed in the rubric below are the required scripts and what you should learn from them to succeed.

Formatting	• <u>Goal</u>: Format your reproduced results to expand on the findings, diving deeper • 1 page PDF write up ○ Includes your findings, experience with the data and analysis, and what you would change about the project. ○ Connect back to the motivation ○ Submit to Canvas
data_cleaning_visualizations.ipynb	• <u>Goal</u>: Clean and prepare the timeseries data for analysis • Load and explore the necessary data to examine the structure, columns, and basic statistics. • Convert date columns to datetime format and sort chronologically • Exploratory Analysis: Analyzes average pollutant levels by area, identifies top/bottom areas, and checks for duplicates • Data Filtering: Creates filtered datasets for: ○ PM2.5 (Fine particles) pollutant only ○ NYC Community Districts (CD) only ○ Combined CD + PM2.5 dataset ○ Visualization: Generates line plot of PM2.5 trends over time and bar chart of top 5 highest PM2.5 community districts
performed_analysis.ipynb	• <u>Goal</u>: Compare STL (Seasonal and Trend decomposition using LOESS) components of Fine Particles across NYC Community Districts grouped by population level. • <u>Research Question</u>: Do high-population and low-population CDs show different patterns in PM 2.5 time series decomposition? • Run STL decomposition on all CD time series for PM 2.5 only • Split CDs into two groups based on population (median split) • Extract trend slope, seasonal amplitude, and residual RMSE for each CD • Compare metrics between groups using Mann-Whitney U test (nonparametric)
References	• Include any additional references used in IEEE format

Acknowledgements: Special thanks to Henry Allen (UVA student) for inspiration for this rubric. This structure is pulled from Javier Rasero's DS4002 Spring 2026 course where he credits [Streifer & Palmer \(2020\)](#).